CS 251
Program 07

Main topics: Composition - Aggregation
Inheritance

Program Specification:

You are to design and implement a set of exactly five classes which can be used to model the most basic feature of an orginization.

- `Person` - has only a name.
- `Member` - extends Person to include a unique id.
- `Leader` - extends Member to include what year of a 2 year term of office they are in.
- `Org`:
  - A Person can only be the Leader of a single Org.
  - A Person can be a Member of multiple Orgs.
  - An Org has a name.
  - An Org has a single Leader.
  - An Org may have many Members.
  - An Org can generate a String representation of the list of its' Leader and all its' Members.
- `Driver` - A class which must fully test all of the above classes' functionality by minimally constructing two unique Orgs - with unique Leaders, and multiple Members - at least one of which is a Member of multiple Orgs.

Rules and Requirements:

- When designing your classes you must use Inheritance, Composition and Aggregation correctly to "Enforce" the defining characteristics above.
- For each class (other than Driver):
  - An appropriate default and specifying constructor must be included. In derived classes, the corresponding *super* constructor must be utilized.
  - All access of instance variables (within each class), by the other instance methods **must** be made via the classes' accessor(s) and/or mutator(s).

Submission:

1. Use your web browser to open:

   `https://uwm.edu`

2. Select [Current Students] from the top menu bar
3. Select [Canvas] from the drop down menu

4. Login to Canvas
5. Click on the COMPSCI 251 block
6. Click on Assignments
7. Click on Program 07 in the left center of the current window
8. Click the **Submit Assignment** button in the right top of the current window
9. Click the **Browse** button in the left center top of the current window
10. Use the *File Upload* pop-up window to find the file you wish to submit
11. Click on this file name in the right panel of the *File Upload* pop-up window
12. Click the **Open** button in the *File Upload* pop-up window
13. Click the **Add** button in the bottom right top of the *Submit a File* pop-up window
14. Click the **Submit Assignment** button in the left bottom of the current window